Prashant Govindarajan

prashant.govindarajan@mila.quebec | gprashant17.github.io | Google Scholar | gprashant17 | LinkedIn | X

Summary

Fifth-year PhD candidate at Mila (advisor: Sarath Chandar) working on **LLMs and reinforcement learning for scientific discovery**, spanning text-to-3D CAD generation and crystal and material discovery. Summer 2026 at Google building agentic LLM systems; research collaborations with Ansys and Intel Labs. Hands-on with LLM training and infrastructure: supervised and RL post-training of LLMs, distributed multi-GPU training on HPC clusters, and evaluation pipelines. First-author publications in *TMLR* and *Digital Discovery* and at multiple NeurIPS/ICML/ICLR workshops, including **CADmium**, a 170k-example text-to-CAD dataset with fine-tuned code LLMs, and **CrystalGym**, an open-source RL environment for crystal design with DFT-computed rewards.

Education

Mila – Quebec AI Institute and Polytechnique Montréal, Montréal, Canada *Aug 2022 – Present*
PhD in Computer Engineering (expected Aug 2027). Advisor: Sarath Chandar

Indian Institute of Technology Madras, Chennai, India *Jul 2017 – Jul 2022*
B.S. Biological Sciences and M.Tech. Data Science (dual degree); Minor in Computational Biology *CGPA: 9.31/10*

Relevant coursework: Reinforcement Learning, Geometry and Generative Models, Representation Learning, Robot Learning, Parallel Scientific Computing, Algorithmic Approaches to Computational Biology

Research Experience

Generative models for text-to-3D CAD *Jun 2024 – Present*
Mila, in collaboration with Ansys (Jay Pathak)
- Led **CADmium**: built a dataset of **170k+** CAD models with human-like text descriptions via a GPT-4.1 annotation pipeline, and fine-tuned code LLMs (1.5B–14B parameters) to generate JSON CAD sequences from text. Published at **TMLR** 2026 [Paper, Code, Website].
- Introduced geometric and topological evaluation metrics for generated CAD (sphericity, mean curvature, Euler characteristic); released dataset, models, and code on Hugging Face and GitHub.
- Currently extending the approach to LLM-driven *editing* of existing CAD sequences.

Generative and RL models for crystalline materials design *Aug 2022 – Present*
Mila, in collaboration with Intel Labs (Santiago Miret, Mariano Phielipp)
- Built **CrystalGym**, the first open-source online RL environment for property-conditioned crystal design in which rewards (band gap, bulk modulus, density) are computed *directly by DFT*; benchmarked value- and policy-based RL algorithms and formalized a new class of RL problems with slow, expensive reward signals. **Spotlight**, AI4Mat @ ICLR 2025 [Paper, Code].
- Developed offline RL (conservative Q-learning) and behavioral-cloning policies that generate crystals conditioned on target band gap from DFT-labelled data, and analyzed learning under noisy DFT rewards. *Digital Discovery* 2024 [Paper, Code]; MoML 2023, AI4Mat @ NeurIPS 2023 and 2024.
- **CrysTune**: RL fine-tuning of LLMs on symmetry-aware Wyckoff representations for property-conditioned crystal generation. AI for Science @ ICML 2026 [Paper].
- Mentored an intern on benchmarking machine-learning interatomic potentials for crystal structure relaxation against Quantum ESPRESSO DFT evaluation. AI for Science @ NeurIPS 2025.

Master's thesis: Graph generative models for binding site-specific molecule generation *Aug 2021 – Jun 2022*
IIT Madras; advisors Balaraman Ravindran, Karthik Raman [Thesis, Poster]
- Designed graph VAEs for target-conditioned drug-like molecule generation; RNN/LSTM decoders for node and edge generation with reduced ordering dependence.

Earlier research in computational biology and genomics *2019 – 2021*
Université de Montréal (Brian Wilhelm), Yale University (Hongyu Zhao), Indian Institute of Science (Debnath Pal)
- Drug–gene response analysis in AML cells, deep generative models (VAEs, scVI) for single-cell RNA-seq, and somatic mutation calling in oral cancer RNA-seq.

Industry Experience

Software Engineering PhD Intern, Google *May 2026 – Aug 2026*
Google Cloud, Global Infrastructure team; Bengaluru, India
- Built an **agentic LLM system** (Google Agent Development Kit) for conversational semantic discovery across heterogeneous internal sources (search, code, and data), designed to be used by multiple teams within Google Cloud's global infrastructure organization.
- Responsible for problem framing, agent design and tool integration, NLP components, and software engineering on an open-ended problem; designed and ran a systematic evaluation of agent behavior (retrieval quality, tool use, failure modes) to guide iteration.
- Delivered a working end-to-end prototype and demo to the team, with a documented roadmap of research-level extensions; handed off for backend integration.

Publications (Google Scholar)

Journal and TMLR

- **Govindarajan, P.***, Baldelli, D.*, Pathak, J., Fournier, Q., Chandar, S. "CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design." ***Transactions on Machine Learning Research***, 2026. (*equal contribution)
- **Govindarajan, P.**, Miret, S., Rector-Brooks, J., Phielipp, M., Rajendran, J., Chandar, S. "Learning Conditional Policies for Crystal Design Using Offline Reinforcement Learning." ***Digital Discovery*** (RSC), 2024.

Selected workshop papers (peer-reviewed)

- **Govindarajan, P.**, Reymond, M., Clavaud, A., Phielipp, M., Miret, S., Chandar, S. "CrystalGym: A New Benchmark for Materials Discovery Using Reinforcement Learning." AI4Mat Workshop, *ICLR 2025* (**Spotlight**); arXiv:2509.23156.
- **Govindarajan, P.**, Das, K., Chandar, S. "CrysTune: Crystal Generation via Fine-Tuning of Large Language Models on Wyckoff Representations." AI for Science Workshop, *ICML 2026*.
- Woo, K., **Govindarajan, P.**, Chandar, S. "Benchmarking Machine Learning Potentials for Crystal Structure Relaxation." AI for Science Workshop, *NeurIPS 2025*.
- **Govindarajan, P.**, Reymond, M., Miret, S., Phielipp, M., Chandar, S. "Crystal Design Amidst Noisy DFT Signals: A Reinforcement Learning Approach." AI4Mat Workshop, *NeurIPS 2024*.
- **Govindarajan, P.**, Miret, S., Rector-Brooks, J., Phielipp, M., Rajendran, J., Chandar, S. "Behavioral Cloning for Crystal Design." ML4Materials Workshop, *ICLR 2023*.

Technical Skills

Generative modeling and ML: PyTorch, TensorFlow, JAX; autoregressive LLMs and code LLMs; diffusion/score-based models and VAEs for 3D and graph-structured data; LLM post-training (SFT and RL-based); offline and online RL; geometric deep learning and GNNs

Chemistry and materials: DFT (Quantum ESPRESSO), pymatgen, ASE, machine-learning interatomic potentials, RDKit

GPU and HPC: C/C++ and parallel programming (OpenMP, MPI, OpenACC); distributed multi-GPU training on HPC clusters (FSDP, DeepSpeed)

Agents and software: Google ADK, LLM agent evaluation, semantic search and vector databases, Python, R, MATLAB, Git

Awards and Fellowships

- **FRQNT Doctoral Research Scholarship** (2026) – Fonds de recherche du Québec, merit-based
- **PBEEE Merit Scholarship for International Students** (2025) – Fonds de recherche du Québec, merit-based
- **Khorana Program for Scholars** (2020) – Dept. of Biotechnology, Govt. of India; U.S. research fellowship (program cancelled due to COVID-19)
- **INSPIRE Scholar** – Dept. of Science and Technology, Govt. of India
- **Summer schools:** Sciencepreneurship (EPFL, 2024); accepted to EEML 2026, Oxford ML 2024, and Amii AI Week 2022 (all declined)

Teaching, Competitions, and Service

- **Teaching assistant:** Machine Learning (Fall 2025, $\sim$ 100 students) and Reinforcement Learning (Fall 2023), Polytechnique Montréal; Reinforcement Learning (Spring 2022) and DSA for Biology (Fall 2021), IIT Madras. Computer Science Instructor, Chandar Lab High School Internship Program.
- **Organizing:** MoML 2023 and MoML 2024 (Molecular ML conference at Mila); co-organizer of the AI for Materials reading group at Mila.
- **Reviewing:** ICLR, TMLR; AI for Science (NeurIPS 2025, ICML 2026); AI4Mat (NeurIPS 2023, 2024; Vienna 2024); MoML 2023–25; REALM @ EMNLP 2026; Deployable AI @ AAAI 2023.
- **Competitions:** First prize, start-up pitch competition, Sciencepreneurship Summer School (EPFL, 2024); winning team, MIT COVID-19 Challenge; finalist, NYAS Tracking Coronavirus challenge.
- **Outreach:** Volunteer, Graduate Application Assistance for Underrepresented Students in AI.
- **Community:** Information Security Committee, Mila; social events organizer, Chandar Lab; Ultimate Frisbee (NSO), sponsorship coordinator for Shaastra 2019 (technical festival), and coordinator of the Analytics Club (Center for Innovation), IIT Madras.

Selected Course Projects

- **Symmetry-aware score-based generative model for crystal structures** (Geometry and Generative Models, McGill, 2022) – incorporated crystal symmetry as an inductive bias in a diffusion model for crystal generation.
- **Transformer for generating drug-like molecules from gene expression signatures** (Algorithmic Approaches to Computational Biology, IIT Madras, 2020) – *de novo* molecule generation inducing target transcriptomic profiles; poster at MLCSB COSI, ISMB 2022 [Poster, Video, Report].
- **GPU parallelization of canonical polyadic tensor decomposition** (Parallel Scientific Computing, IIT Madras, 2021) – CPU/GPU implementations with OpenMP and OpenACC [Report, Code].